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E-Catalog



“ Beyond Boundaries, We Command Control ”

Product Catalog



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Company Profile

EPBOX ENGINEERING is a trusted innovator in the design and manufacturing of intelligent control panels and industrial automation solutions. Guided by our motto “**Beyond Boundaries, We Command Control**,” we deliver systems that empower our clients with reliability, precision, and adaptability in the most demanding environments.

Founded on 14 March 2017 and strategically headquartered in Singapore, **EPBOX ENGINEERING** has grown into a recognized solutions provider for mission-critical industries, including Petrochemical, Oil & Gas, Water Treatment, Buildings, and Infrastructure. Our expertise spans from Low Voltage (LV) Power Distribution Panels to Programmable Logic Controller (PLC) based automation and Supervisory Control And Data Acquisition (SCADA) systems. Each project customized to meet the unique operational requirements of our clients.

Our expertise covers:

- Tailored PLC and SCADA Programming
- Customized automation logic for specific operational needs
- I/O Mapping and P&ID Interpretation
- Accurate translation of engineering schematics into functional systems
- Redundant Control System Design
- High reliability architectures for mission-critical operations
- Custom Human Machine Interfaces (HMI)
- User friendly operator panels for efficient control and monitoring
- Secure Industrial Networking
- Cybersecure connectivity for industrial environments
- Compliance with Offshore, Marine, and Data Centres Standards
- Adhering to strict international regulations
- Explosion-Proof Panel Capabilities
- Expertise in **Ex-d**, **Ex-e**, and **Ex-p** designs for hazardous locations

Our Commitment :

- To deliver high-integrity control systems that meet international standards.
- To support clients from concept through commissioning.
- To innovate across electrical, software, and integration layers.

Sectors We Serve :

- Oil & Gas (FPSO, Refinery, Onshore)
- Power & Renewable Energy
- Marine & Offshore Systems
- Industrial Automation
- Data Centres & Critical Infrastructure
- Water Treatment & Building Automation



Our Vision & Mission

Vision	Mission
To be the one stop-solution for industrial control and automation, delivering integrated services from panel design to system commissioning, empowering global transformation.	To revolutionize industries worldwide with cutting-edge panel solutions, driving progress, growth, and connectivity toward a sustainable future.



What We Offer



- **Panel Engineering:** LV Panels, MCC, PLC, and HMI solutions, from design to Factory Acceptance Testing (FAT) and full documentation.
- **PLC & SCADA Programming:** Tailored automation with Siemens, Mitsubishi, Omron, and integrated SCADA / remote access for real time monitoring and control.
- **ATEX & Offshore Compliance:** Certified solutions designed for hazardous areas and offshore environments.
- **Custom HMI Design:** User-friendly operator interfaces that enhance productivity and efficiency.
- **Industrial Networking:** Secure and robust communication systems for critical operations.
- **Retrofit & Upgrade:** Modernization of legacy systems with seamless integration.
- **System Design Support:** Assistance throughout the project lifecycle, from concept to commissioning.

System Integration

EPBOX ENGINEERING specializes in delivering end-to-end automation solutions, seamlessly integrating SCADA platforms, Building Management Systems (BMS), and industrial automation into a unified control environment.

From offshore platforms and process plants to data centers and smart buildings, we design systems that provide real-time visibility, predictive control, and operational efficiency.

SCADA, BMS & Automation Systems

- ☐ BMS Solutions: Integrated HVAC, power, fire detection, safety, and security systems.
- ☐ Automation Systems: Full PLC integration with intelligent control logic.
- ☐ Communication Protocols: Expertise in Modbus, Profibus, Profinet, Ethernet/IP, OPC UA, and BACnet.
- ☐ Reliability & Redundancy: Implementation of redundant servers and robust failover strategies to ensure uptime.
- ☐ Cybersecurity: Secure industrial networking featuring advanced access control and data protection.
- ☐ Custom HMI Design: Intuitive, user-friendly operator interfaces designed for maximum efficiency and control.



Our Key Products



PLC Control Panel

Reliable automation solution for mission-critical operations, integrating PLCs, I/O modules, power supplies, and communication systems into a unified system.

Features:

- Real-time control & monitoring for optimal efficiency.
- Suitable for manufacturing, oil & gas, water treatment, and power generation.

Explosion Proof Panels

(Ex-d, Ex-e, Ex-p)

Designed for safe and reliable operation in hazardous areas with flammable gases, vapors, or dust.

Features:

- Certified for Ex-d (Flameproof), Ex-e (Increased Safety), Ex-p (Pressurized) under international standards.
- Prevents internal ignition from spreading to the external environment.
- Ideal for oil & gas, petrochemical, and chemical processing industries.



Power Distribution Panel (LV)

The central system for safely distributing electrical energy across facilities, ensuring consistent and reliable power supply.

Features:

- Divides main feed into protected subsidiary circuits.
- Equipped with premium busbars, breakers, and safety devices for long-term reliability.

Motor Control Centre (MCC LV)

A centralized solution for managing and protecting electric motors, delivering safe, reliable, and efficient operation.

Features:

- Integration of motor starters, VFDs, and automation hardware in a single enclosure.
- Comprehensive protection against overloads, short circuits, and phase faults.
- Streamlined maintenance with extended equipment lifespan.



Local Control Panel (LCP)

Provides direct, on-site manual control for specific equipment, ensuring quick access and safe intervention.

Features:

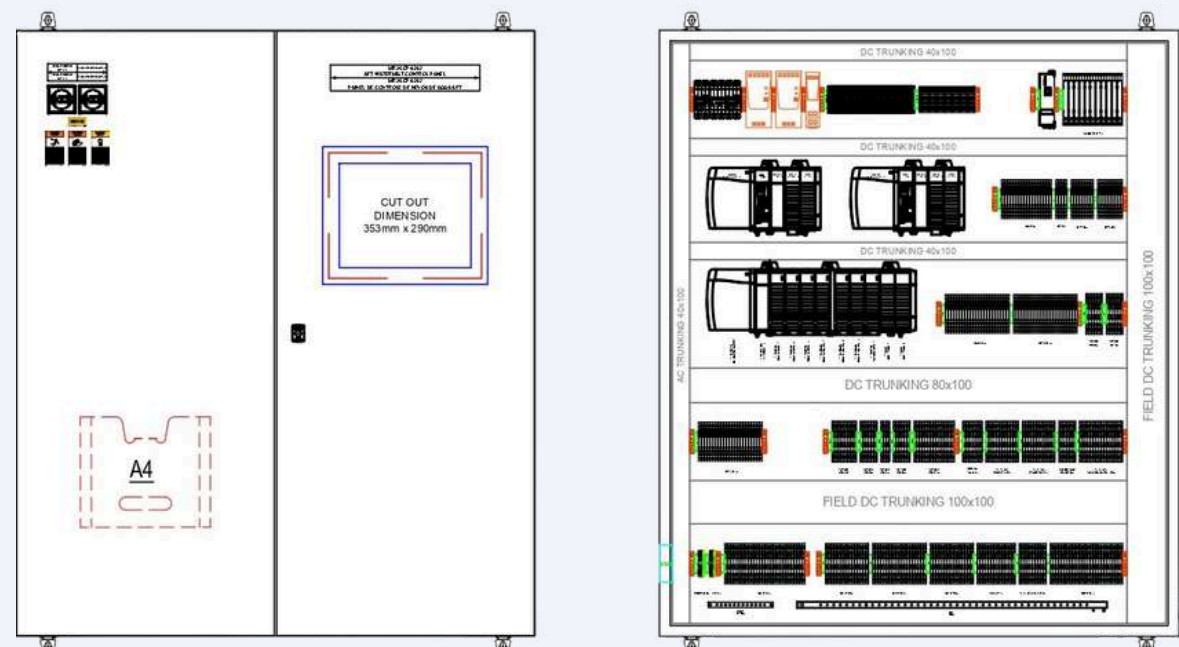
- Manual start/stop and monitoring near the equipment.
- Intuitive controls, clear indicators, and emergency stop for operator safety.
- Ideal for pumps, fans, conveyors, and standalone process units.

PLC Control Panel

EPBOX ENGINEERING PLC Control Panel is engineered as the core automation unit for industrial processes. By integrating programmable logic controllers with I/O modules, protective devices, and operator interfaces, the panel ensures precise, safe, and continuous control of critical systems.

Key Features

- **Standards-Based Design:** Engineered in compliance with key international and local standards (IEC, CP5, NR10), tailored to specific project requirements.
- **Broad PLC Compatibility:** Seamless integration with all leading PLC platforms, including Siemens, Allen-Bradley, Schneider Electric, Omron, and Mitsubishi.
- **Multi-Protocol Communication:** Fluent in major industrial protocols such as Modbus, Profibus, Profinet, Ethernet/IP, and OPC UA.
- **Modular & Scalable Architecture:** Features a flexible I/O design that easily accommodates both digital and analog signals as system requirements evolve.
- **Durable Enclosures:** Available with enclosures rated up to IP66, providing dust-tight and water-resistant protection for harsh environments.
- **Optimized for Safety & Maintenance:** A clean internal layout designed for safe operation, straightforward troubleshooting, and long-term maintainability.



Project Technical Specifications:

Parameter	Specification
Operating Voltage	110 – 690 V AC / 24 V DC (PLC module dependent)
Frequency	50 / 60 Hz
Controller Type	Siemens, Allen-Bradley, Schneider, Omron, Mitsubishi and etc
Communication Protocols	Modbus, Profibus, Ethernet/IP, Profinet, OPC UA
I/O Capacity	Scalable (digital & analog inputs/outputs)
Enclosure Protection	Protection from IP42 to IP66 and beyond
Standards Compliance	Available in custom configurations

Explosion Proof Panels

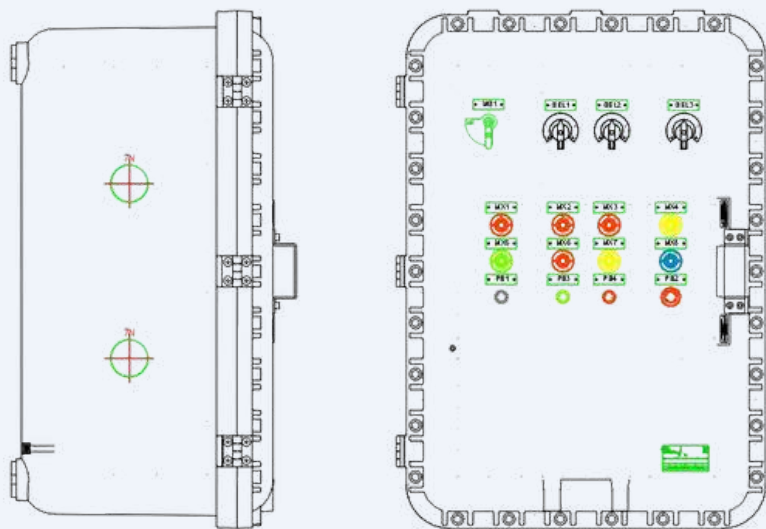
(Ex-d, Ex-e, Ex-p)

EPBOX ENGINEERING provides explosion-proof panels engineered and certified for use in hazardous environments where flammable gases, vapors, or dusts are present. Every panel we supply is designed in accordance with ATEX / IECEx, IEC, NR10, and CP5, ensuring absolute reliability and operational safety.

Available in Ex-d (Flameproof), Ex-e (Increased Safety), and Ex-p (Pressurized/Purged) types. Our solutions are tailored to project requirements and built to withstand the most demanding industrial and offshore conditions. From MCC and PLC control systems to process monitoring panels, we deliver solutions that guarantee continuous operation and maximum protection.

Key Features

- **Certified Compliance:** Engineered to meet and exceed key international standards, including ATEX, IECEx, IEC 61439, NR10, and CP5.
- **Protection:** Features enclosures rated up to IP66, providing superior protection against dust and water ingress in harsh industrial and offshore environments.
- **Flexible Configurations:** Fully customizable for a wide range of applications, including Motor Control Centres (MCCs), PLC systems, and specialized process control units.
- **Explosion Proof Types:** Ex-d (Flameproof), Ex-e (Increased Safety), Ex-p (Pressurized)
- **Engineered for Reliability:** Designed for safe, long-term performance in demanding sectors such as oil & gas, petrochemical, and critical infrastructure.



Project Technical Specifications:

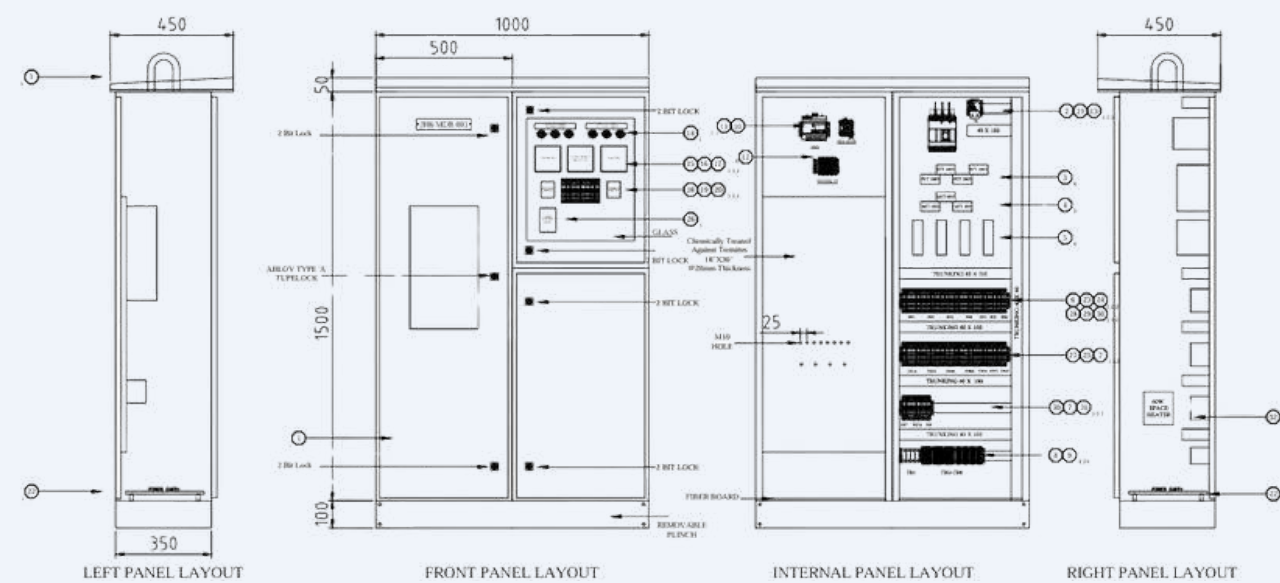
Parameter	Specification
Protection Types	Ex-d, Ex-e, Ex-p
Operating Voltage	110 – 690 V AC / 24 V DC
Frequency	50 / 60 Hz
Enclosure Protection	Protection from IP42 to IP66 and beyond
Material	Aluminum or Stainless Steel 316
Temperature Class	T4 – T6
Standards Compliance	IECex, ATEX or Compatible certification

Power Distribution Panel

EPBOX ENGINEERING Power Distribution Panel (PDP) provides safe, efficient, and reliable power distribution for industrial plants, offshore facilities, and utilities. Engineered to meet or exceed key industry standards-including IEC, CP5, and NR10-each panel is built for continuous operation in the most demanding environments.

Key Features

- High-Capacity Busbars: Rated up to 2000 A, with higher capacity ratings engineered to client specifications.
- High Short Circuit Rating: Engineered to safely withstand short circuit currents up to 100 kA for maximum safety.
- Robust Enclosure Protection: Available with enclosure ratings up to IP66 / NEMA 4X, ensuring superior protection in harsh indoor and outdoor environments.
- Advanced Protection Devices: Integrates high-quality MCCBs (Molded Case Circuit Breakers) and ACBs (Air Circuit Breakers), with selective coordination available to ensure system reliability.
- Smart Monitoring & Control: Equipped with digital metering, protective relays, and SCADA-ready communication protocols for seamless system integration.
- Future-Ready Design: Features a modular and scalable architecture, allowing for simple and cost-effective future expansion.



Project Technical Specifications:

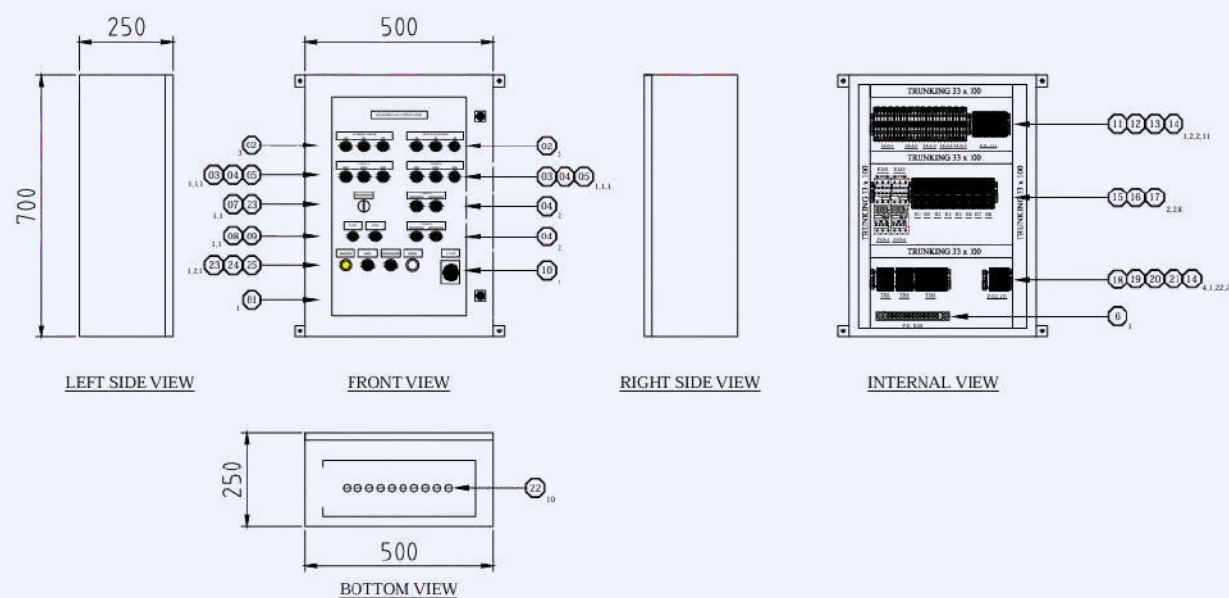
Parameter	Specification
Rated Voltage	220V - 690V AC
Rated Current	Up to 2000A (customizable)
Frequency	50 / 60 Hz
Protection	Protection from IP42 to IP66 and beyond
Busbar Material	Copper/Aluminium, IEC standard compliant
Circuit Breakers	MCCB, ACB, MCB (as per design)
Installation Type	Floor mounted / Wall mounted

Motor Control Centre (MCC LV)

EPBOX ENGINEERING LV MCC delivers provides centralized, safe, and efficient control of low voltage motors for industrial and offshore facilities. Engineered in full compliance with key standards like IEC, CP5, and NR10, our MCCs integrate robust protection, modular construction, and intelligent monitoring to ensure reliable performance in mission-critical applications.

Key Features

- **High Current Rating:** Standard ratings up to 2000 A, with higher capacities engineered to meet specific client requirements.
- **Superior Fault Protection:** High short circuit withstand capacity, tested up to 100 kA, to ensure maximum operational safety.
- **Enhanced Operator Safety:** Designed with internal segregation up to Form 4b to minimize risk to personnel during operation and maintenance.
- **Versatile Motor Control:** Supports a full range of starting methods, including Direct-On-Line (DOL), Star-Delta, Soft Starter, and VFD schemes.
- **Durable Enclosures:** Available with enclosure protection up to IP66, ensuring long-term reliability in offshore and other harsh environments.
- **Intelligent System Integration:** Features optional communication protocols for seamless integration with plant-wide SCADA and Building Management Systems (BMS).



Project Technical Specifications:

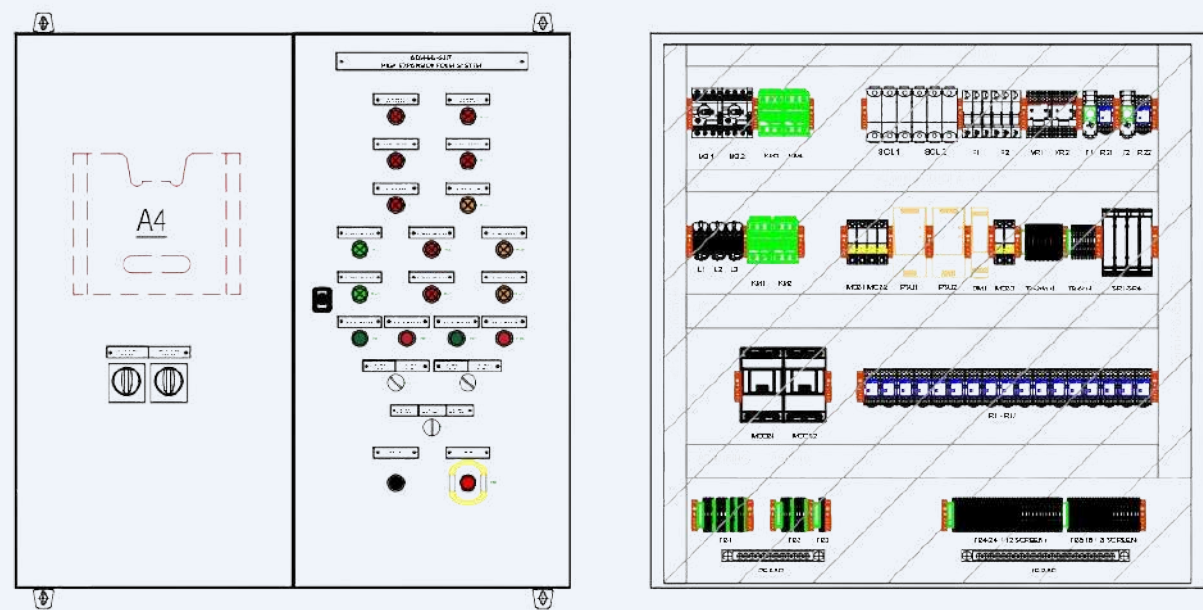
Parameter	Specification
Operating Voltage	380 – 690 V AC
Frequency	50 / 60 Hz
Current Capacity	up to 2000 A
Protection	Protection from IP42 to IP66 and beyond
Busbar System	Copper/Aluminium, IEC standard compliant
Motor Starter Types	DOL, Star Delta, Soft Starter, VFD
International Standards	Available in custom configurations

Local Control Panel (LCP)

EPBOX ENGINEERING Local Control Panel (LCP) provides operators with safe, direct, and on-site manual control of field equipment. Designed for reliability in industrial and process environments, LCPs ensure immediate access for monitoring and intervention, independent of central control systems. Engineered to meet stringent safety and performance standards, our LCPs integrate intuitive controls, protection features, and durable construction for mission-critical applications.

Key Features

- **Direct Field Control:** Enables start/stop and equipment monitoring directly at site, ensuring fast operator response.
- **Operator Safety:** Built with emergency stop, lockable switches, and clear status indicators.
- **Versatile Configurations:** Available in standard or customized layouts to support pumps, fans, conveyors, and other field equipment.
- **Rugged Enclosures:** Designed for harsh environments with protection up to IP66 for outdoor or industrial use.
- **System Integration:** Compatible with SCADA, PLC, or central DCS systems for seamless plant-wide control.



Project Technical Specifications:

Parameter	Specification
Operating Voltage	110 – 690 V AC
Frequency	50 / 60 Hz
Protection	Protection from IP42 to IP66 and beyond
Control Options	Push buttons, selector switches, HMI, emergency stop
Materials	Available in custom configurations
Integration	PLC, SCADA, or DCS compatible
International Standards	Available in custom configurations

Technical Capability

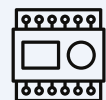
At **EPBOX ENGINEERING**, we deliver end-to-end automation solutions from conceptual design to commissioning ensuring every system meets the highest standards of safety, performance, and reliability.

Our expertise spans explosion proof panels, PLC & SCADA integration, and building management systems (BMS), giving clients complete control and visibility over their critical operations.

Our Capabilities



Panel Engineering.
(LV, MCC, PLC, HMI solution)



PLC, BMS & SCADA Programming and Integration.



ATEX & Offshore Compliance.



System Integration and Custom HMI Design and Industrial Networking.



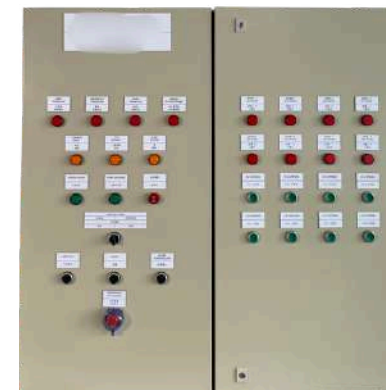
System Troubleshooting & Lifecycle Support.

Why It Matters

With **EPBOX ENGINEERING**, clients gain more than panels they gain intelligent, connected automation systems. Our technical capabilities ensure safe, reliable, and future ready control solutions, trusted across oil & gas, utilities, industrial plants, and mission critical facilities worldwide.

Project Reference

Every project we deliver represents our commitment to engineering excellence, operational reliability, and client success. From offshore control panels to factory automation systems, our work speaks for itself.



FPSO Fire & Gas (F&G) Control Panel.

Explosion-proof control panel designed to meet the strict safety and operational demands of FPSO vessels.

- ATEX-certified components for hazardous areas.
- Seamless integration with vessel SCADA systems.
- Full lifecycle support, including offshore commissioning with EPC partners.

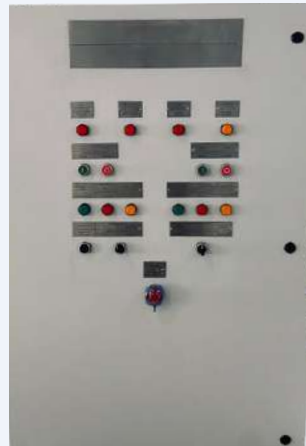


Smart Factory Retrofit PLC & HMI Integration.

Upgrade legacy machines with modern PLC and HMI systems-boosting productivity, enabling real-time data, and delivering smart factory performance without the cost of new equipment.

- Zero-Downtime Retrofits: Executed during planned maintenance for uninterrupted production.
- Multi-Brand Expertise: Integration of Siemens, Allen-Bradley, Omron, and more.
- Centralized SCADA: Unified monitoring and control across the production floor.

Project Reference



Cleanroom Panel & Power Management System

Power Distribution Panels (LV) and local control cabinets engineered for cleanroom critical operations.

- Space saving and compact layout.
- Surge & noise protection for sensitive equipment.
- Seamless BMS integration.



Data Centre Generator Control Panel

PLC based generator control panel designed for automated backup power switching in mission critical data centre operations.

- Intelligent failover logic programming.
- Load sharing with real-time monitoring.
- Redundant wiring for maximum reliability.



ABB / Qatar RAS ABU Desalination Plant

Comprehensive engineering, design, and supply of electrical panels to support large-scale desalination operations at the RAS ABU Plant in Qatar.

- Customized panel design meeting client specifications.
- Integrated protection and control systems for reliable plant operations.
- Built for durability and efficiency in demanding desalination environments.

Why Do Clients Choose Us



Certified Engineering

A skilled team with proven experience in offshore, industrial, and marine-grade systems.



Turnkey Delivery

Complete project execution, from I/O mapping and P&ID interpretation to full SCADA deployment.



Compliant Systems

Expertise in designing systems for strictly regulated environments requiring IECEx, ATEX, ABS, and DNV certifications.



Responsive Support

Rapid technical assistance and remote troubleshooting capabilities to minimize downtime.



EPBOX ENGINEERING

Your Trusted Engineering Partner.

Company Highlights

Notes



No.	Project Name	Size	EPC/Main-Con/Producer
1	Petronas Inert Gas Fire Suppresion System	1 Panel (Ex-D)	Deluge/Petronas
2	Google Data Center Collaboration with Rovysis	26 Panels	Google
3	Analyzer House HVAC system (Siemens)	6 panels (Ex-D)	Siemens
4	450kVA GenSet controls (Aerogates)	9 Panels	Aerogates
5	RAS Abu Fontas Desalination Project	2 Panels	ABB/Qatar
6	Infineum ESD system	1 Panels	Jacobs/Infineum
7	MCC Control System	3 Panels	Cyclelect/BW-Offshore
8	A-Star Data Center	6 Panels	A-Star
9	Sembcorp Gas Recovery and Scrubber System	1 Panels	Solution
10	Socar Ammonia and Ura Complex Heat Tracing Panels	2 Panels	Greentech/H&E
11	Meiden PUB Ulu Pandan Desalination Water Project	10 Panels	Meiden/PUB
12	Shell ESD System	3 Panels	Rockwell/Shell
13	FM200 F&G System	1 Panels	Deluge/Oceanmight
14	Inergen GAS Fire Fighting System	2 Panels	Wilsafe/Barossa FPSO
15	Watermist System	3 Panels	Semco/MODEC FPSO
16	High Expansion Foam System	1 Panels	Semco/Yinson
17	CEMS Analyzer Exhaust Emission PLC Control Panel	2 Panels	Exxon Mobile
18	Fire Fighting Water Sprinkle System	10 Panels	Majestic Ferry
19	BMC-33 FPSO Complete Fire Fighting Control Panels	10 Panels	Modec FPSO
20	UARU NOVEC Fire Fighting System Control Panel	1 Panels	Modec/Exxon Mobile
21	Mero 3 Watermist System Siemens Safety PLC Redundancy System	2 Panels	MIRC/Petrobas